

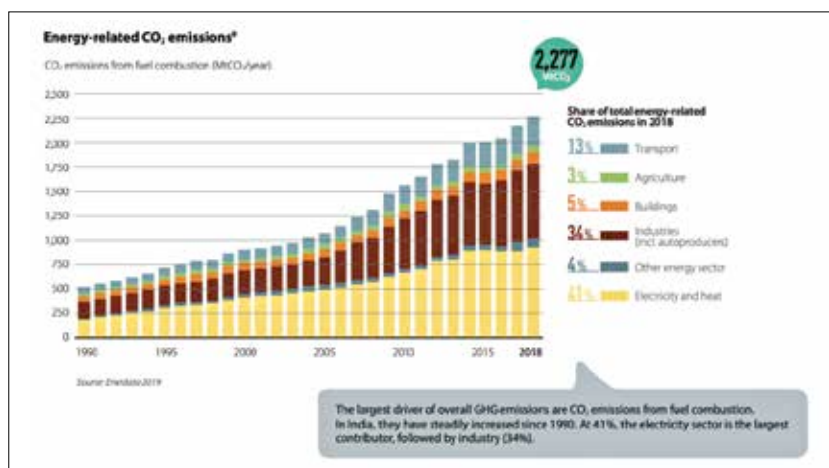
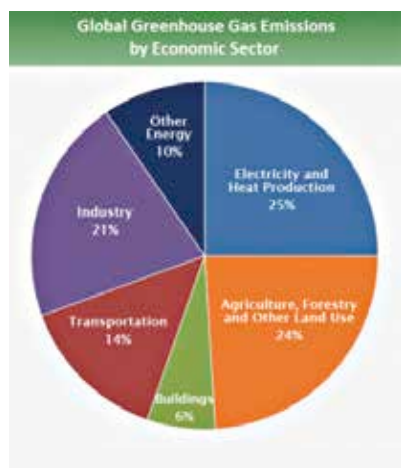


Fig. 1 (left) Global and Indian (right) energy related GHG emissions from different sectors.

and people are making 20-25% IRR on the initial capital expenditure. Since Chennai has a huge solar irradiance of 5.08 kWh/m²/day, solar power has become one of the cheapest sources of energy there.

However, a larger question is why are renewables not scalable? This is because neither solar nor wind energy can produce energy throughout the day. For example, the US state of California has been at the forefront of solar power, and aims to produce 50% of its electricity from renewables by 2030! Since solar only works in the day and household demand is more at night, there's a mismatch between supply and demand. This is called the "Duck effect". One has to use non-solar energy sources or batteries along with grid power to account for this. Another major challenge is the amount of space required for grid scale solar and wind installations requiring huge displacement of local wildlife and clearing out of vegetation, which doesn't make sense when that's the same environment you're trying to protect.

Renewable technologies need a strong and stable grid power setup with low or nil GHG emission. In this context, nuclear energy can step in as the perfect partner for renewables.

Advanced nuclear reactors based on 'Thorium' are now being allocated as the great green hope of clean energy production, producing less waste and more energy than Ura-

nium. 1 Ton of Thorium produces energy equivalent to 200 Ton of Uranium and around 3.5 Million Tons of Coal. What's even better is that Thorium is abundantly present in the world. Just to put that in perspective, a dice size piece of Thorium can generate enough energy to fulfil the lifetime requirements of an average human being. Other major safety advantages of Thorium-based reactors are that they don't require highly pressurized water, they are melt-down-proof, have no weapons-grade by-products and can even consume legacy plutonium stockpiles.

India has been working on a very ambitious yet important three-stage nuclear power production program to utilize the country's vast reserves of Thorium-232. India has the largest reserves of Thorium in the world. However, over the past decade this program has lost a lot of steam, due to cost overruns, technical issues, vested interests and push away from nuclear due to the recent Fukushima-Daiichi nuclear disaster of 2011.

Bill Gates chairs a Company called "Terrapower" which is working on innovation in "Thorium based nuclear energy", which is an effective, zero emission technology. Bill Gates states that "Nuclear is ideal for dealing with climate change, because it is the only carbon-free, scalable energy source that's available 24 hours a day. The problems with today's reactors,

such as the risk of accidents, can be solved through innovation".

A very good comparison of nuclear and renewable energy lies in this example: France generates 72% of its electricity from nuclear and just 6% from solar and wind, whereas Germany generated 44.6% of its electricity through renewables. Due to unreliability of solar and wind sources, Germany had to spend 27% more on things like transmission lines from distant farms to cities which increases the cost of electricity. Thus, electricity in France today is 41% cheaper than Germany. Also, since solar farms need to be re-built every 15-25 years where nuclear plants last for more than 80 years and very less materials are used in the building nuclear plants (920 tonnes/TWh) as compared to solar plants (~ 16500 tonnes/TWh), it seems that the French model of electricity generation is a better approach than that of Germany that relies heavily on renewables. An important statistic is that only 1/10th of the Carbon pollution in France is attributed to its electricity production!

A right mix of Thorium based nuclear power with renewables could be our best solution in saving the world while sustaining human growth on this planet. For a progressive intelligent civilization, it is important to move towards more and more concentrated sources of energy generation, and Thorium-232 seems to be that 'magic bullet' we need. 